

Q1. Hill climbing Kya hai?

A local search algorithm jo har step mein best neighbor choose karta hai.

Q2. Hill climbing mein "hill" Kya represent karta hai?

Objective/fitness function ki value.

Q3. Simple hill climbing kab ruk jaata hai?

Jab koi better neighbor na mile (local maxima).

Q4. Hill climbing ka main drawback Kya hai?

Local maxima mein fas jaata hai, global maxima nahi milta.

Q5. Hill climbing greedy hai ya not?

Haan, yeh greedy algorithm

hai.

Q6. Hill climbing mein initial state kya hoti hai?

Randomly ya manually chosen starting point.

Q7. Simple us Steepest Hill Climbing mein farq?

Simple pehla better neighbor leta hai, Steepest sabse best leta hai.

Q8. Hill climbing complete algorithm hai?

Nahi, yeh complete nahi hai.

Q9. Hill climbing optimal hai?

Nahi, optimal solution guarantee nahi karta.

Q10. Plateau problem kya hai hill climbing

mein?

Jab sab neighbors ki value equal ho, progress ruk jaati hai.

Q11. Hill climbing Python mein kaise start hota hai?

```
python current = initial_state
```

Q12. Neighbor ki value check karne ka code?

```
python if neighbor_value > current_value:  
current = neighbor
```

Q13. Simple hill climbing ka basic loop?

```
python while True:
```

```
neighbor = get_neighbor(current)
```

```
if f(neighbor) > f(current):
```

```
current = neighbor
```

```
else:
```

```
break
```

Q14. Objective function example (maximize x^2)?

```
python def f(x):
```

```
return x * x
```

Q15. Random restart hill climbing kyu use

Karte hain?

Local maxima se bachne ke liye alag jagah se dobara start Karte hain.

Q16. Hill climbing mein termination condition kya hoti hai?

Jab current state se koi improvement possible na ho.

Q17. Hill climbing ka time complexity?

$O(\infty)$ worst case, depends on landscape.

Q18. Simulated Annealing hill climbing se better kyu hai?

Yeh kabhi kabhi worse solution bhi accept karta hai, local maxima se bachne ke liye.

Q19. Hill climbing ka real life

```

import random

# Objective function (we want to maximize this)
def objective(x):
    return -(x - 3)**2 + 9    # Maximum at x = 3

# Hill Climbing function
def hill_climbing():
    current = random.randint(-10, 10)    # random start
    step_size = 1

    while True:
        neighbor = current + random.choice([-step_size, step_size])

        # If neighbor is better, move there
        if objective(neighbor) > objective(current):
            current = neighbor
        else:
            break    # stop if no improvement

    return current, objective(current)

# Run algorithm
best_x, best_value = hill_climbing()

print("Best Solution x:", best_x)
print("Best Value:", best_value)

```

```

... Best x = 1.9543, f(x) = 12.9979

```

example?

GPS shortest path dhundhna, game Aal move selection.

Q20. Google Colab mein hill climbing run karne ke steps?